297mm

Front

210mm

pharmaniaga®

Flumazenil 0.1mg/mL Injection

PRODUCT DESCRIPTION
A clear, colourless solution.

The compatible infusion solutions for intravenous use are as follows:
- Sodium Chloride 0.9% Solution
- Dextrose 5% Solution

The solution should be clear, colourless solution when diluted.
If not required immediately, the diluted solution may be stored up to 24 hours at temperature <30ºC after mixing with the compatible solutions.

COMPOSITION
Each ampoule contains Flumazenil 0.50mg.

PHARMACODYNAMICS
Pharmacotherapeutic group: All other therapeutic products, Antidotes.

ATC code: V03A B25

Flumazenil, an imidazobenzodiazepine, is a benzodiazepine antagonist which, by competitive interaction, blocks the effects of substances acting via the benzodiazepine-receptor. Neutralisation of paradoxal reactions of benzodiazepines has been reported.

According to experiments in animals, the effects of substances, which are not acting via the benzodiazepine-receptor (like barbiturates, GABA-mimetics and adenosine-receptor agonists), are not blocked by flumazenil. Non-benzodiazepine-agonists, like cyclopyrrolones (zopiclone) and triazolopyridazines, are blocked by flumazenil. The hypnosedative effects of benzodiazepines are blocked rapidly (within 12 minutes) after intravenous administration.

Depending on the difference in elimination time between agonist and antagonist, the effect can recur after several hours. Flumazenil has possibly a slight agonistic, anticonvulsive effect. Flumazenil caused withdrawal, including convulsions in animals receiving long-term flumazenil treatment.

PHARMACOKINETICS
Distribution
Flumazenil is a lipophilic weak base. Flumazenil is bound for approximately 50% to plasma proteins, from which two thirds are bound to albumin. Flumazenil is extensively divided over extra vascular space. During the distribution phase plasma concentration of flumazenil decreases with a half-life of 4-15 minutes. The distribution volume under steady-state conditions (Vss) is 0.9-1.1 L/kg.

Metabolism
Flumazenil is mainly eliminated through hepatic metabolism. The carboxylic acid metabolite was shown in plasma (in free form) and in urine (in free and conjugated form) to be the most important metabolite.
In pharmacological tests this metabolite has proved to be inactive as benzodiazepine agonist or antagonist.

Elimination
Almost no unchanged flumazenil is excreted in the urine. This indicates a complete metabolic degradation of the active substance in the body. Radiolabelled medicinal product is completely eliminated within 72 hours, with 90 to 95% of the radioactivity appearing in the urine and 5 to 10% in the faeces. Elimination is rapid, as is shown by the short half-life of 40 to 80 minutes. The total plasma clearance of flumazenil is 0.8 to 1.0 L/hour/kg and can almost completely be attributed to hepatic metabolism.
The pharmacokinetics of flumazenil is dose-proportional within the therapeutic dose-range and up to 100 mg.

The intake of food during the intravenous infusion of flumazenil results in an increase of 50% of the clearance probably due to postprandial increase in liver perfusion.
Pharmacokinetics in special patient groups

Elderly
The pharmacokinetics of flumazenil in elderly is not different from that in young adults.

Patients with impaired hepatic function
In patients with a moderately to severely impaired liver function the half-life of flumazenil is increased (increase of 70-210%) and the total clearance is lower (between 57 and 74%) compared to normal healthy volunteers.

Patients with impaired renal function
Pharmacokinetics of flumazenil is not different in patients with impaired renal function or patients undergoing haemodialysis compared to normal healthy volunteers.

Paediatric population
In children above one year of age, the half-life elimination is shorter and the variability is higher than in adults, approximately of 40 min with a range of 20 to 75 min. Clearance and volume of distribution, by kg of body weight are the same as in adults.

INDICATION
Pharmaniaga Flumazenil 0.1mg/mL Injection is indicated for the complete or partial reversal of the central sedative effects of benzodiazepines. It may therefore be used in anaesthesia and in intensive care in the following situations:

In anaesthesia
- Termination of hypnosedative effects in general anaesthesia induced and/or maintained with benzodiazepines in hospitalised patients.
- Reversal of benzodiazepine sedation in short-term diagnostic and therapeutic procedures in ambulatory patients and hospitalised patients.

In intensive care situations
- For diagnosis of intoxication with benzodiazepines or to rule out such intoxication.
- As a diagnostic measure in unconsciousness of unknown origin to differentiate between involvement of benzodiazepines, other drugs or brain damage.
- For specific reversal of the central effects of benzodiazepines in drug overdose (return to spontaneous respiration and consciousness in order to render intubation unnecessary or allow extubation).

RECOMMENDED DOSAGE
Pharmaniaga Flumazenil 0.1mg/mL Injection is to be administered intravenously by an anaesthetist or experienced physician.

Pharmaniaga Flumazenil 0.1mg/mL Injection may also be administered as an infusion.

When Flumazenil Injection is to be used as an infusion, it must be diluted prior infusion. Pharmaniaga Flumazenil 0.1mg/mL Injection should only be diluted with sodium chloride 0.9% solution and dextrose 5% solution. Intravenous infusion solution should be used immediately and be discarded after 24hours.

Pharmaniaga Flumazenil 0.1mg/mL Injection may also be used concomitantly with other resuscitative measures. Dosage should be titrated for the required effect. The action of some benzodiazepines may exceed the effect of flumazenil, repeated doses may be required if sedation recurs following awakening.

Adults
Anaesthesia
The recommended starting dose is 0.2 mg administered intravenously over 15 seconds. If the required level of consciousness is not obtained within 60 seconds, a further dose of 0.1 mg can be injected and repeated at 60-second intervals, up to a maximum dose of 1.0 mg. The usual dose required lies between 0.3 and 0.6 mg, but may vary depending on the patient's characteristics, the dose and duration of effect of benzodiazepine used.

Intensive care
The recommended starting dose is 0.3 mg administered intravenously. If the required level of consciousness is not obtained within 60 seconds, a further dose can be injected and repeated at 60-second intervals, up to a total dose of 2 mg or until the patient awakes. If drowsiness recurs, an intravenous infusion of 0.1 - 0.4 mg/h may be useful. The rate of infusion should be adjusted individually to achieve the desired level of consciousness.
If a significant improvement in consciousness and respiratory function is not obtained after repeated doses of flumazenil, a non-benzodiazepine aetiology must be assumed.

To avoid withdrawal symptoms in patients treated for a long period of time with high doses of benzodiazepines in the intensive care unit, the dosage of flumazenil has to be titrated individually and the injection has to be administered slowly. If symptoms of overstimulation arise during the use of flumazenil, then diazepam or midazolam may be administered by slow intravenous injection.

Special Populations

Elderly
In the absence of data on the use of flumazenil in elderly patients, it should be noted that this population is generally more sensitive to the effects of medicinal products and should be treated with due caution.

Patients with renal or hepatic impairment
Since flumazenil is primarily metabolised in the liver, careful titration of dosage is recommended in patients with impaired hepatic function. No dosage adjustments are required in patients with renal impairment.

Paediatric population

Children above 1 year of age
For the reversal of conscious sedation induced by benzodiazepines in children older than 1 year, the recommended starting dose is 0.01 mg/kg (up to 0.2 mg) administered intravenously over a period of 15 seconds. If, after an additional waiting period of 45 seconds, the required level of consciousness is not obtained, a follow-up injection of 0.01 mg/kg (up to 0.2 mg) may be administered and if necessary, repeated at 60-second intervals (up to a maximum of 4 times) to a maximum dose of 0.05 mg/kg or 1 mg, depending on which is the lowest dose. The dose should be adjusted to the patient's response.

Children under the age of 1 year
There are insufficient data on the use of flumazenil in children younger than 1 year.
Therefore flumazenil should only be administered in children younger than 1 year if the potential benefits to the patient outweigh the possible risk.

Instruction for use/method of dilution
When flumazenil is to be used in infusion, it must be diluted prior to infusion. Flumazenil should only be diluted with sodium chloride (0.9%) solution and 5% dextrose solution. 50 mL of Pharmaniaga Flumazenil 0.1 mg/mL Injection (equivalent to 5 mg of Flumazenil) may be diluted into 50 mL infusion beg to get the concentration of 0.05 mg/mL.Compatibility between flumazenil and other solutions for injection has not been established.

Intravenous infusion solution should be used immediately and be discarded after 24hours.

ROUTE OF ADMINISTRATION
Parenteral
[For IV use only.]

CONTRAINDICATIONS
Flumazenil is contraindicated in patients:
- With known hypersensitivity to flumazenil or to any of the excipients.
- Patients receiving benzodiazepines for control of a potentially life-threatening condition (e.g. control of intracranial pressure or status epilepticus).
- In mixed intoxications with benzodiazepines and tricylic and/ or tetracyclic antidepressants, the toxicity of the antidepressants can be masked by protective benzodiazepine effects.
  In the presence of autonomic (anticholinergic), neurological (motor abnormalities) or cardiovascular symptoms of severe intoxication with tricyclics/tetracyclics, flumazenil should not be used to reverse the benzodiazepine effect.

WARNINGS AND PRECAUTIONS
Use in children for other indications than reversal of conscious sedation is not recommended as no controlled studies are available. Until sufficient data are available, flumazenil should only be administered to children below the age of 1 year if the risks to the patient (especially in the case of accidental overdose) have been weighed up against the benefits of the treatment.

- Elimination may be delayed in patients with hepatic impairment.
- The patient should be monitored for an adequate period of time based on the dose and duration of effect of the benzodiazepine employed (ECG, pulse, oximetry, patient alertness and other vital signs such as heart rate, respiratory rate and blood pressure).
- The antagonistic effect of flumazenil is specific to benzodiazepines; an effect is therefore not to be expected if the 'nonawakening' is caused by other substances.
- When used in anaesthesiology at the end of surgery, flumazenil should not be given until the effects of peripheral muscle relaxants have been fully reversed.
- As the action of flumazenil is usually shorter than that of benzodiazepines and sedation may possibly recur the patient should remain closely monitored, preferably in the intensive care unit, until the effect of flumazenil has presumably worn off.
- In high-risk patients, the benefits of benzodiazepine-induced sedation should be weighed against the risks of rapid awakening. In patients (e.g. with cardiac problems) maintenance of a certain level of sedation may be preferable to being fully awake.

attn	customer Pharmaniaga LifeScience Sdn Bhd		date 28.04.2021
1 colours	process colours: Black		Please Chop & Sign For Approval
size description	297 x 210mm material 60gm Simili Flumazenil Insert (2003023)- Front		
FocusPrint SDN BHD	t: 603-8766 6030 f: 603-8766 6033 (Factory) website: www.focusprint.com.my		
artwork prepared by: cynthia yap	email: graphic@focusprint.info (Graphic Dept)		date:

ARTWORK LOG

Revision no.	Date	Reason for Change
01	05.05.2020	- New insert received.
02	28.04.2021	- Add ® at Logo, amend co. no and Revision Date.

IMPORTANT ! Please note that this colour print is for visual purpose only. Output colour might differ in actual production.

297mm

Back

210mm

- Rapid injection of flumazenil should be avoided. In patients with high-dose and/or long-term exposure to benzodiazepines ending at any time within the weeks preceding flumazenil administration, rapid injection of doses equal to or higher than 1 mg has led to withdrawal symptoms, including palpitations, agitation, anxiety, emotional labiality as well as mild confusion and sensory distortions.
- In patients suffering from preoperative anxiety or having a history of chronic or episodic anxiety the dosage of flumazenil should be adjusted carefully.
- After major surgery, postoperative pain must be taken into account and it may be preferable to keep the patient lightly sedated.
- In patients treated for long periods with high doses of benzodiazepines, the advantages of the use of flumazenil should be weighed against the risk of withdrawal symptoms. If withdrawal symptoms occur despite careful dosing individually titrated low doses of benzodiazepines (diazepam or midazolam) should be given by slow intravenous injection.
- Because of the potential for resedation and respiratory depression children previously sedated with midazolam should be monitored at least 2 hours after flumazenil administration. In case of other sedating benzodiazepines, the monitoring time must be adjusted according to their expected duration.
- The use of the antagonist is not recommended in patients with epilepsy, who have been treated with benzodiazepines for a prolonged period of time. Although flumazenil has some intrinsic antiepileptic effects, the abrupt antagonizing effect can cause convulsions in patients with epilepsy.
- In patients with severe brain injury (and/or instable intracranial pressure) receiving flumazenil – to reverse the effects of benzodiazepines – an increased intracranial pressure may develop.
- Particular caution is necessary when using flumazenil in cases of mixed drug overdose. In particular in the case of an intoxication with benzodiazepines and cyclic antidepressants, certain toxic effects such as convulsions and cardiac arrhythmias, which are caused by these antidepressants but which emerge less readily on concomitant administration with benzodiazepines, are exacerbated on administration of flumazenil.
- Patients who have received Flumazenil for the reversal of benzodiazepine effects should be monitored for resedation, respiratory depression or other residual benzodiazepine effects for an appropriate period based on the dose and duration of effect of the benzodiazepine employed. Because patients with underlying hepatic impairment may experience delayed effects as described above, an extended observation period may be required.
- Flumazenil is not recommended for the treatment of benzodiazepine-dependence or for the treatment of long-term benzodiazepine-abstinence-syndromes.
- Panic attacks have been reported after the use of flumazenil in patients with a history of panic disorder.
- Due to the increased frequency of benzodiazepines tolerance and dependence in patients with alcoholism and other drug dependencies, flumazenil should be used with caution in its population.
- Removed statement (not to be included in PI)

Effects On Ability To Drive And Use Machines
Patients who have received flumazenil to reverse the effects of benzodiazepine sedation should be warned not to drive, to operate machinery or to engage in other activities demanding physical or mental exertion for at least 24 hours, since the effect of the benzodiazepine may return.

INTERACTIONS WITH OTHER MEDICAMENTS
Flumazenil reverses the central effects of benzodiazepines by means of competitive interaction at receptor level: the effects of non-benzodiazepine agonists acting via the benzodiazepine receptor, such as zopiclone, triazolopyridazine and others, are also antagonised by flumazenil. However, flumazenil does not block the effect of medicines that do not operate via this route. Interaction with other central nervous system depressants has not been observed. Particular caution is necessary when using flumazenil in cases of accidental overdose since the toxic effects of other psychotropic medicinal products (especially tricyclic antidepressants) taken concurrently may increase with the subsidence of the benzodiazepine effect.

No change in the pharmacokinetics of flumazenil has been observed in combination with the benzodiazepines such as midazolam, flunitrazepam and lormetazepam. Flumazenil does not affect the pharmacokinetics of these benzodiazepines.

There is no pharmacokinetic interaction between ethanol and flumazenil.

INCOMPATIBILITIES
This medicinal product must not be mixed with other medicinal products except for those mentioned in section 'Product Description and Recommended Dosage'.

PREGNANCY AND LACTATION
Pregnancy
Although studies in animals have not shown evidence of embryo toxicity or teratogenicity, the possible risk to humans caused by flumazenil during pregnancy has not been determined. Therefore, flumazenil should only be used during pregnancy if the possible benefit to the patient outweighs the potential risks for the foetus.

Lactation
It is not known whether this flumazenil is excreted in breast milk. For this reason, breast-feeding should be interrupted for 24 hours when flumazenil is used during lactation. Emergency use of flumazenil during lactation is not contraindicated.

SIDE EFFECTS
Any side effects associated with Flumazenil usually subside rapidly without the need for special treatment.
Immune systems disorders
Common: Allergic reactions
Rare: Severe hypersensitivity reactions (including anaphylaxis)

Psychiatric disorders
Common: Anxiety*, emotional lability, insomnia, somnolence
Uncommon: Fear
Unknown: Withdrawal symptoms, (e.g. agitation, anxiety, emotional lability, confusion, sensory distortions, tachycardia, dizziness, sweating), following rapid injection of doses of 1 mg or more in patients with high-dose and/or long-term exposure to benzodiazepines ending at any time within the weeks preceding flumazenil administration panic attacks (in patients with a history of panic reactions); abnormal crying, agitation, aggressive reactions (the side effect profile in children is generally similar to that in adults. When Flumazenil has been used for the reversal of conscious sedation, abnormal crying, agitation and aggressive reactions have been reported).

Nervous system disorders
Common: Vertigo, headache, agitation*, tremor, dry mouth, hyperventilation, speech disorder, paresthesia
Uncommon: Convulsions (in patients suffering epilepsy or severe hepatic insufficiency, mainly after long-term treatment with benzodiazepines or multiple medicinal products abuse).

Eye disorders
Common: Diplopia, strabismus, lacrimation increased

Ear and labyrinth disorders
Uncommon: Abnormal hearing

Cardiac disorders
Common: Palpitations*
Uncommon: Tachycardia or bradycardia, extrasystole

Vascular disorders
Common: Flushing, hypotension, orthostatic hypotension, transient increased blood pressure (on awakening)

Respiratory, thoracic and mediastinal disorders
Uncommon: Dyspnoea, cough, nasal congestion, chest pain

Gastrointestinal disorders
Very common: Nausea (during anaesthesia)
Common: Vomiting (during anaesthesia), hiccup

Skin and subcutaneous tissue disorders
Common: Sweating

General disorders and administration site conditions
Common: Injection site pain
Uncommon: Shivering
Rare: Severe hypersensitivity reactions (including anaphylaxis)

*: following rapid injection, generally did not require treatment

OVERDOSE AND TREATMENT
There is very limited experience of acute overdose in humans with Flumazenil.

There is no specific antidote for overdose with Flumazenil. Treatment of an overdose with Flumazenil should consist of general supportive measures including monitoring of vital signs and observation of the clinical status of the patient.

Even when given at doses exceeding those recommended, no symptoms of overdosage were observed. For withdrawal symptoms attributable to the agonist, see Warnings and Precautions.

STORAGE CONDITIONS
Store below 30°C. Protect from light.

The solution should be clear, colourless solution when diluted.
If not required immediately, the diluted solution may be stored up to 24 hours at temperature <30ºC after mixing with the compatible solutions.

SHELF LIFE
36 months.

DOSAGE FORMS AND PACKAGING AVAILABLE
- 10 x 5mL ampoule (clear)

PRODUCT REGISTRATION HOLDER /MANUFACTURER
Pharmaniaga Lifescience Sdn Bhd (198201002939)
Lot 7, Jalan PPU 3,
Taman Perindustrian Puchong Utama,
47100 Puchong, Selangor Darul Ehsan, Malaysia

2003023

Revision Date: 28-April-2021

attn	customer Pharmaniaga LifeScience Sdn Bhd		date 28.04.2021
1 colours	process colours: Black		Please Chop & Sign For Approval
size description	297 x 210mm Flumazenil Insert (2003023)- Back	material 60gm Simili	
FocusPrint SDN BHD	t: 603-8766 6030 f: 603-8766 6033 (Factory) website: www.focusprint.com.my		
artwork prepared by: cynthia yap	email: graphic@focusprint.info (Graphic Dept)		date:

ARTWORK LOG

Revision no.	Date	Reason for Change
01	05.05.2020	- New insert received.
02	28.04.2021	- Add ® at Logo, amend co. no and Revision Date.

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